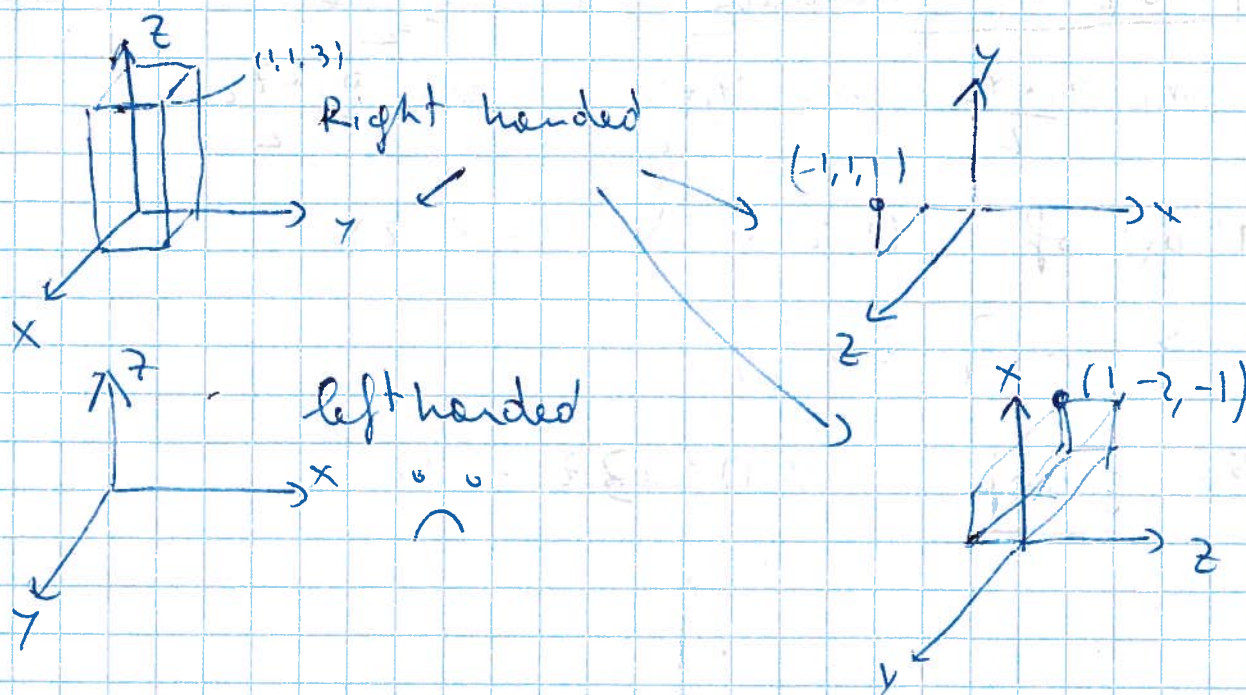


## 11.1 Cartesian coords in 3-space



Distance formula  $|P_1 P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Eg  $\left| \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} \right| = \sqrt{(-1-1)^2 + (3-2)^2 + (-2-1)^2} = \sqrt{4+1+9} = \sqrt{14}$

Eg Sphere of radius 3 centered at (1, 3, 2)

$$(x-1)^2 + (y-3)^2 + (z-2)^2 = 3^2 = 9$$

Eg Find centre & radius of  $x^2 + 2x + y^2 - 6y + z^2 + 8z - 10 = 0$

$$(x+1)^2 - 1 + (y-3)^2 - 9 + (z+4)^2 - 16 - 10 = 0$$

$$(x+1)^2 + (y-3)^2 + (z+4)^2 = 36$$

$$R = 6$$

$$C = (-1, 3, -4)$$



Midpoint formula  $M = \frac{P_1 + P_2}{2}$

$$m_1 = \frac{x_1 + x_2}{2}$$

$$m_2 = \frac{y_1 + y_2}{2}$$

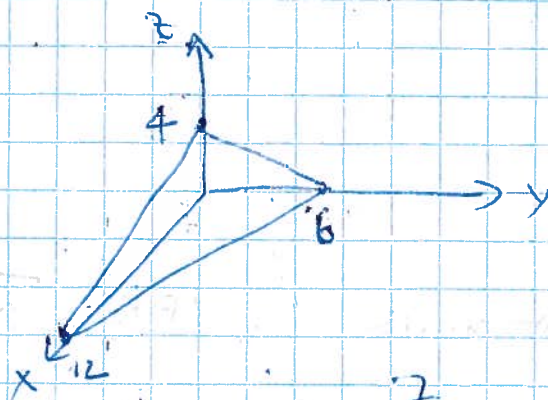
$$m_3 = \frac{z_1 + z_2}{2}$$

Midpt of  $(1, 2, 3)$  &  $(-4, 5, 2)$

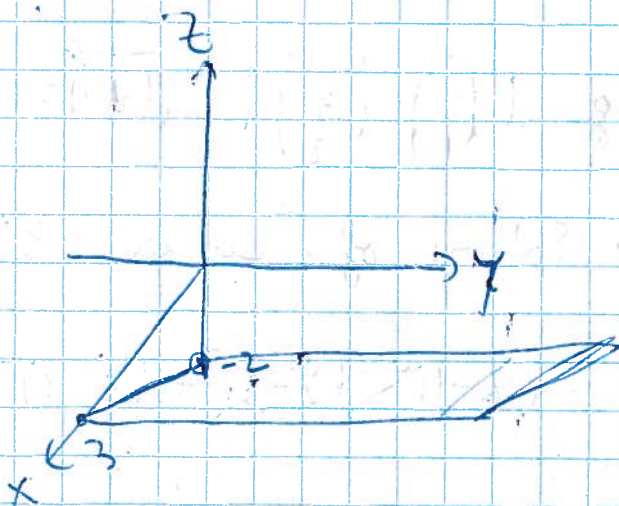
is  $(-\frac{3}{2}, \frac{7}{2}, \frac{5}{2})$

Ex graph

$$x + 2y + 3z = 12$$



$$2x - 3z = 6$$

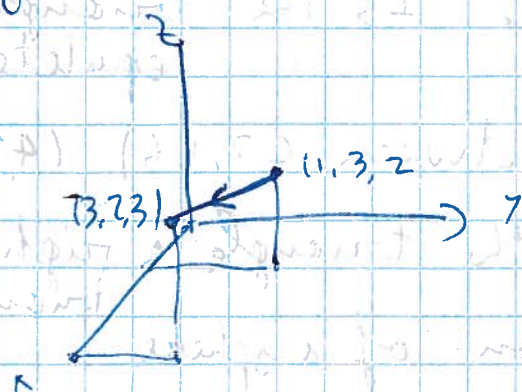


∴ No  $y$  in the equation  $\Rightarrow$  parallel to the  $y$  axis



CURVE  $x = f(t)$   $y = g(t)$   $z = h(t)$   $a \leq t \leq b$

eg  $x = 1+2t$   $y = 3-t$   $z = 2+t$   $0 \leq t \leq 1$



direction  $\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

$x' = 2$   
derivative  $y' = -1$   
 $z' = 1$

$$L = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} dt$$

$$\int_0^1 \sqrt{2^2 + (-1)^2 + 1^2} dt = \int_0^1 \sqrt{6} dt = \sqrt{6}t \Big|_0^1$$

$$= \sqrt{6}(1-0) = \sqrt{6}$$

distance  $\sqrt{(1-3)^2 + (3-2)^2 + (2-3)^2} = \sqrt{4+1+1} = \sqrt{6} \checkmark$

$x = \cos t$   
 $y = \sin t$   
 $z = t$   
 $0 \leq t \leq 2\pi$



$x' = -\sin t$   
 $y' = \cos t$   
 $z' = 1$

$$L = \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t + (1)^2} dt = \int_0^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2}$$

Ex 1-39 ODD

note that  
distance  $\left| \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2\pi \end{pmatrix} \right| = \sqrt{(2\pi)^2} = 2\pi$   
 $\uparrow$   
 $2\pi\sqrt{2}$



Exercises: 1) Plot  $(1, 2, 3)$ ,  $(2, 0, 1)$ ,  $(-1, 3, 4)$

2) Find the distances between the points

$(4, 5, 3)$ ,  $(1, 7, 4)$ ,  $(2, 4, 6)$ . Is the triangle equilateral?

3) Find the distances between  $(2, 1, 6)$ ,  $(4, 7, 9)$

and  $(8, 5, -6)$ . Is the triangle a right angle triangle?

4) Write the equation of a sphere with center  $(2, -1, -5)$  & radius 4.

5) Find center and radius of the sphere

$$x^2 + 2x + y^2 - 3y + z^2 - z = 0$$

6) Sketch the planes  $x + 3y - z = 6$ ,

$$3x + 4z = 12, \quad 3x - 4y + 2z = 12$$

7) Find the arc length of

a)  $x = t$   $y = t$   $z = 2t$   $0 \leq t \leq 2$

b)  $x = t^{3/2}$   $y = t^{3/2}$   $z = t$   $2 \leq t \leq 4$

c)  $x = t^2$   $y = (4/3)t^{3/2}$   $z = t$   $0 \leq t \leq 8$

8) Find the equation of a sphere

with diameter  ~~$(-2, 3, 6)$   $(4, -1, 5)$~~

given by the segment  $(-2, 3, 6)$   $(4, -1, 5)$